

AIM:

To develop a web application that validates user inputs using JavaScript and stores data in a database using Servlet/JSP for an online product comparison system.

ALGORITHM:

1. Start the program
2. Create HTML Form
 - a. Design a form to collect product details:
 - i. Product Name
 - ii. Brand
 - iii. Price
 - iv. Category
3. Apply JavaScript Validation
 - a. Validate that all fields are not empty
 - b. Check that brand contains only alphabets
 - c. Ensure price is numeric and greater than zero
 - d. Ensure a category is selected
4. Submit the Form
 - a. If validation is successful, submit the form using POST method to the servlet
5. Receive Data in Servlet
 - a. Retrieve form values using request.getParameter()
6. Load JDBC Driver
 - a. Load MySQL driver using Class.forName()
7. Establish Database Connection
 - a. Connect to MySQL using DriverManager.getConnection()
8. Prepare SQL Query
 - a. Create an INSERT query using PreparedStatement
9. Execute Query
 - a. Store product details in the database
10. Send Response
11. Display success message to user
12. Stop the program

CODE:**product.html**

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>

    <title>TechScope Product Entry</title>

    <link rel="stylesheet" href="style.css">

    <script src="validate.js"></script>

</head>

<body>

<div class="card">

    <h1>Add Product</h1>

    <form action="ProductServlet" method="post" onsubmit="return validateForm()">

        <input type="text" id="name" name="name" placeholder="Product Name"><br><br>

        <input type="text" id="brand" name="brand" placeholder="Brand"><br><br>

        <input type="text" id="price" name="price" placeholder="Price"><br><br>

        <select id="category" name="category">

            <option value="">Select Category</option>

            <option value="Mobile">Mobile</option>

            <option value="Laptop">Laptop</option>

        </select><br><br>

        <button type="submit">Submit</button>

    </form>

</div>

</body>

</html>
```

validate.js

```
function validateForm() {
```

```
let name = document.getElementById("name").value;

let brand = document.getElementById("brand").value;

let price = document.getElementById("price").value;

let category = document.getElementById("category").value;


if (name === "") {

    alert("Product name is required");

    return false;

}


let brandPattern = /^[A-Za-z ]+$/;

if (!brand.match(brandPattern)) {

    alert("Brand must contain only alphabets");

    return false;

}


if (isNaN(price) || price <= 0) {

    alert("Enter a valid price");

    return false;

}


if (category === "") {

    alert("Select a category");

    return false;

}


return true;

}
```

ProductServlet.java

```
package com.techscope;
```

```

import jakarta.servlet.*;

import jakarta.servlet.http.*;

import java.io.*;

import jakarta.servlet.annotation.WebServlet;

import java.sql.*;

@WebServlet("/ProductServlet")

public class ProductServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse response throws
ServletException, IOException {

        String name = request.getParameter("name");

        String brand = request.getParameter("brand");

        int price = Integer.parseInt(request.getParameter("price"));

        String category = request.getParameter("category");

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/techscope","root","zaara/25");

            PreparedStatement ps = con.prepareStatement(

                "INSERT INTO product(name, brand, price, category) VALUES (?, ?, ?, ?)"

            );

            ps.setString(1, name);

            ps.setString(2, brand);

            ps.setInt(3, price);

            ps.setString(4, category);

            ps.executeUpdate();

            out.println("<html><head><link rel='stylesheet' href='style.css'></head <body>");

```

```

        out.println("<div class='card'>");

        out.println("<h1>Product Stored Successfully </h1>");

        out.println("<p>Name: " + name + "</p>");

        out.println("<p>Brand: " + brand + "</p>");

        out.println("<p>Price: ₹" + price + "</p>");

        out.println("<p>Category: " + category + "</p>");

        out.println("</div></body></html>");

        con.close();

    }

    catch (Exception e) {

        out.println("<h2>Error: " + e.getMessage() + "</h2>");

    }

}

}

```

OUTPUT:



Name	Brand	Price	Category
M51	Samsung	₹28750	Mobile
Iphone 14	Apple	₹33199	Mobile

RESULT:

Thus, The form inputs are validated using JavaScript, and valid data is successfully stored in the database using a servlet.